

Leo Shaw

leos@andrew.cmu.edu

408.769.9693

[LinkedIn](#)

[Github](#)

Education

Carnegie Mellon University

Master of Science in Artificial Intelligence Engineering - Mechanical Engineering

Pittsburgh, PA

Dec 2026

Rutgers University

Bachelor of Science in Mechanical Engineering; Bachelor of Arts in Computer Science

New Brunswick, NJ

Dec 2024

Skills

Programming: Python, C++, Java, PyTorch, TensorFlow, Keras, OpenCV, YOLO, OCR, VLMs, RAG

Data/Infrastructure: PostgreSQL, Docker, FastAPI, JSONL

Design/Hardware: SOLIDWORKS, ANSYS, PLC/HMI integration, circuitry, soldering, 3D printing

Research & Projects

AI Research Engineer

Pittsburgh, PA / Carnegie Mellon University, May 2026 - Present

- Built a multimodal intelligence platform for semiconductor equipment, transforming technical documents, figures, handwritten operator logs, and process telemetry into a unified data layer for AI agents, semantic search, and equipment diagnostics.
- Engineered a scalable perception and ingestion pipeline combining Nemotron OCR, Granite Vision, native document extraction, asynchronous GPU inference, adaptive batching, and automated quality routing to produce traceable, database-ready technical records.
- Developed an evidence-grounded industrial reasoning system using the Laguna SLM, OpenWebUI, and vLLM, combining structured generation, embedding-based clustering, hybrid retrieval, and vector search to generate source-cited answers, failure hypotheses, and recommended diagnostic checks.
- Designed an auditable SLM training-data pipeline with deterministic provenance, document-isolated splits, target-model difficulty probing, semantic evaluation, human-review gates, and reasoning-on/off exports for supervised fine-tuning.

Autonomous Driving Local Planner with Reinforcement Learning

Jan 2026 - May 2026

- Engineered an end-to-end autonomy pipeline in CARLA for highway merge and unprotected left-turn scenarios, combining graph-based routing, PPO policy learning, deterministic control, and route-conditioned state features in a closed-loop simulation stack.
- Built the full ML experimentation workflow in Python with PyTorch and CARLA, including scenario generation, reward shaping, training orchestration, rollout evaluation, and performance benchmarking.
- Improved policy performance across both driving tasks, increasing success rates from 76% to 83% and 44% to 80%, while reducing highway-merge 90th-percentile completion time from 24.5s to 15.25s.
- Structured the project as a reusable AI system rather than a standalone model, with modular components for planning, control, evaluation, and scenario-driven testing.

Real-Time Conveyor-Belt Object Detection and Robotic Sorting

Jan 2026 - May 2026

- Built a real-time AI perception and control pipeline for robotic sorting, using YOLO, OpenCV, homography-based coordinate transforms, IK, and delta-arm motion control to connect visual detections to physical actions.
- Developed the end-to-end system in Python across distributed hardware and software components, integrating edge compute, cloud vision inference, robot control, and timing logic into a closed-loop automation workflow.
- Designed post-detection refinement and motion-compensation logic to improve coordinate accuracy, account for conveyor motion, and reduce perception-to-actuation lag in deployment conditions.
- Achieved up to 99% classification accuracy and approximately 50% end-to-end physical sorting accuracy, using system-level evaluation to identify latency, conveyor-speed variation, and mechanical compliance as the main operational bottlenecks.

Experience

Field Service Engineer I

Santa Clara, CA / Ulvac Technologies, Oct 2024 - Aug 2025

- Rebuilt and installed an upgraded 8-inch etch chuck assembly, integrated a new VAT system, circuit breaker, valve-control updates, and wiring changes to improve tool flexibility, reducing unplanned downtime by 15%, and increasing wafer temperature uniformity.
- Sustained uptime across semiconductor tools across 6 high-volume fabs through preventive maintenance, calibration, and data-driven repairs, increasing uptime by 18%.
- Executed 30+ hardware and software retrofit projects across vacuum, RF, motion-handling, PLC, and HMI subsystems, improving process stability by 22% and extending tool life.
- Led root-cause analysis with engineering and manufacturing teams, implemented corrective actions, and standardized troubleshooting procedures, reducing average fault diagnosis time by 35%.